An Internship Report

Of

C++ Programming

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)

By

Roll Number of Student:- 25SCS1003004157

Name of Student: Yuganshu Kumar Jha

Batch: 2025-29

CANDIDATE'S DECLARATION

I, YUGANSHU KUMAR JHA, do hereby declare that the internship report titled "**C++ PROGRAMMING**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: **YUGANSHU KUMAR JHA**

Roll No.: **25SCS1003004157**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **THIRANEX VIRTUAL INTERNSHIP** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **C++ PROGRAMMING**. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: **YUGANSHU KUMAR JHA**

Roll No.: **25SCS1003004157**

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3. INTERNSHIP COMPLETION CERTIFICATE


Thiranex
SKILL DEVELOPMENT & FUTURE TECH
MSME
Certificate
OF ACHIEVEMENT
This acknowledgment is proudly presented to
Yuganshu Kumar Jha
This is to certify that Yuganshu Kumar Jha has successfully completed an internship in
C++ Programming from 31 Jul 2026 to 30 Aug 2026.
VERIFIED CERTIFICATE
ID:THX-JUL3126-108
Thiranex
Hariharan M
Founder & CEO

Internship Offer Letter

Date: 03 August 2026

TO,

Yuganshu Kumar jha

Intern ID: THX-AUG0326-424

Subject: Offer Letter for Internship in C++ Programming

Dear **Yuganshu Kumar jha,**

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	**Intern - C++ Programming**
Commencement Date	**03 Aug 2026**
Completion Date	**02 Sept 2026**
Work Mode	**Remote / Project-Based**

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.

Thiranex

This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.

www.thiranex.in

MSME

COMPLETED

Due Date: **Immediate Submission**

Submitted

Mandatory: Offer Letter & LinkedIn Post

Download your offer letter, post it on LinkedIn tagging Thiranex, and share the post URL here.

Completed Successfully!

1

COMPLETED

Due Date: **10 Aug 2026**

Submitted on track

Student Management System

To develop a console-based Student Management System in C++ that efficiently manages student records using file handling and menu-driven operations.

Key Features:

- Add, update, delete, and display student information.
- Use file handling to store data persistently.
- Implement menu-driven operations for smooth navigation.

Expected Outcome:

A functional application that allows users to add, update, delete, and display student information with persistent data storage for reliable record management.

Completed Successfully!

2

COMPLETED

Due Date: **17 Aug 2026**

Submitted on track

Bank Management Application

To design and implement a C++ based Bank Management Application that simulates core banking operations using object-oriented programming and file handling.

Key Features:

- Include features like deposit, withdrawal, and balance check.
- Use object-oriented programming concepts.
- Store customer data securely using file management.

Expected Outcome:

A secure and functional banking system capable of performing deposits, withdrawals, and balance inquiries while maintaining persistent customer records.

Completed Successfully!

3

COMPLETED

Due Date: **24 Aug 2026**

Submitted on track

Library Management System

To develop a Library Management System that efficiently manages books, members, and borrowing records using structured or object-oriented programming.

Key Features:

- Implement structures or classes for book and member details.
- Support book issue and return features.
- Include search functionality by title or author.

Expected Outcome:

A functional application that enables book addition, issue and return processing, and search by title or author for streamlined library operations.

Completed Successfully!

4

COMPLETED

Due Date: **31 Aug 2026**

Submitted on track

Mini Game Project (Tic Tac Toe / Snake Game)

To create a console-based mini game in C++ that demonstrates core programming concepts such as loops, arrays, and conditional logic.

Key Features:

- Use loops, arrays, and condition statements for game logic.
- Display the game board dynamically after each move.
- Add win/loss detection and replay options.

Expected Outcome:

An interactive game with dynamic board display, win/loss detection, and replay functionality, showcasing effective implementation of game logic.

Completed Successfully!

AVAILABLE

Share Certificate on LinkedIn

Post your verified certificate on LinkedIn tagging Thiranex.

Internship Processing

Your 30-day internship period completes on **02 Sept 2026**. Certificate download will be unlocked on this date.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 4-week Virtual Internship undertaken under the THIRANEX Virtual Internship Program, in the domain of **"C++ Programming"**. The internship was carried out from July 2026 to August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to bridge the gap between theoretical understanding of Artificial Intelligence and Machine Learning and their real-world, production-grade deployment. Over the course of the programme, the learner was exposed to industry-relevant tools and practices used to take AI/ML models from a research or prototype stage to a fully deployed, monitored, and scalable production system.

The internship followed a structured, week-wise curriculum that combined conceptual learning with hands-on assignments, weekly assessments, project documentation, and a final capstone project. On successful completion of all the requirements, including the final assessment test, a Virtual Internship Certificate was awarded by THIRANEX INDIA.

4.2 Organization Profile

Thiranex India is a software development and technology learning platform dedicated to fostering talent and innovation across various domains. The organization provides virtual internship programs designed to bridge the gap between academic learning and industry requirements by offering hands-on experience in cutting-edge technologies such as Web Development, App Development, Data Science, Cyber Security, and Python/Java programming. Through real-world project assignments, remote learning frameworks, and structured mentorship, Thiranex helps students and aspiring tech professionals strengthen their technical skill sets, build practical portfolios, and enhance their overall employability.

4.3 Problem Statement

The project aims to develop practical C++-based applications that solve simple real-world problems while demonstrating fundamental programming concepts.

The internship tasks focus on creating:

Student Management System

Bank Management Application

Library Management System

Mini Game Project (Tic Tac Toe/ Snake Game)

4.4 Project Objectives

- To develop practical applications using C++ programming.
- To understand and implement C++ fundamentals.
- To use conditional statements and loops for program control.
- To work with String, OOPS, Functions.
- To implement basic Methods for storing and retrieving information.
- To automate repetitive tasks using C++ scripts.
- To develop a simple rule-based Management System using predefined responses.
- To improve problem-solving and logical-thinking skills.
- To create simple, user-friendly command-line applications.

4.5 Scope of the Project

The scope of this project covers the development of simple and practical C++-based applications designed to demonstrate fundamental programming and problem-solving skills. The project includes the implementation of tasks such as a text-based Management System and Game. It involves the use of C++ programming concepts including variables, conditional statements, loops, functions, strings, user input/output. The developed applications are intended for learning and demonstration purposes and operate primarily in a local or command-line environment.

4.6 Technologies and Tools Used

- Programming Language : C++
- Development : C++ IDE/VS Code /IDLE
- Version Control : Git and GitHub
- Data Structure : OOPS,String,Functions
- User Interaction :input() / print()
- Logic : if-else,loops,functions

4.7 System Architecture

The Student Management System follows a **modular, console-based architecture** developed using C++. The system uses a menu-driven interface to allow users to manage student records efficiently. Data is stored persistently using **file handling**, so records remain available even after the program is closed.

Architecture Components

1. **User Interface Layer**
   - Provides a console-based menu.
   - Allows users to select operations such as Add, Update, Delete, Search, and Display.
   - Accepts user input and displays system results.
2. **Application/Business Logic Layer**
   - Processes user requests.
   - Performs operations on student records.
   - Validates student information and manages program flow.
   - Implements functions such as adding, updating, deleting, searching, and displaying records.
3. **Student Data Model**
   - Represents student information using C++ structures/classes.
   - Stores details such as Student ID, Name, Course, Age, Contact, etc.
   - Provides organized access to student records.
4. **File Handling / Data Storage Layer**
   - Stores student records in files such as `.txt` or `.dat`.
   - Reads existing records when required.
   - Updates and deletes records while maintaining persistent storage.

4.8 Methodology

The Student Management System is developed using a **step-by-step modular methodology** in C++. The development process focuses on efficient student record management, persistent data storage, and simple user interaction.

1. Requirement Analysis

Identify the basic requirements of the system, including adding, updating, deleting, searching, and displaying student records.

2. System Design

Design a **menu-driven console application** with separate functions for each major operation. Define the student data structure/class and file storage mechanism.

3. Implementation

Develop the system using **C++ and Object-Oriented Programming (OOP)** concepts. Implement functions for:

- Add Student
- Update Student
- Delete Student
- Search Student
- Display All Students

4. File Handling

Use C++ file handling to store student records permanently. The system reads data from files when required and writes updated information back to the storage file.

5. Input Validation

Validate user inputs such as Student ID and other required details to reduce invalid or duplicate records and ensure reliable data management.

6. Testing

Test each module individually and then test the complete system to verify that all operations work correctly and data is stored and retrieved properly.

7. Deployment and Usage

After successful testing, the application can be compiled and executed as a **console-based C++ application**, allowing users to manage student records through the menu interface.

4.9 Expected Outcomes

The **Student Management System** successfully provides a simple and reliable way to manage student records through a console-based C++ application. The system demonstrates effective use of **Object-Oriented Programming, file handling, and menu-driven operations**.

Key Outcomes

- Successfully adds, updates, deletes, searches, and displays student records.
- Provides **persistent data storage** using file handling.

- Reduces manual effort in maintaining student information.
- Provides a simple and user-friendly console interface.
- Demonstrates practical implementation of **C++ OOP concepts**.
- Maintains student records efficiently for future retrieval.
- Improves accuracy and organization of student data management.

Final Expected Outcome

A functional Student Management System that efficiently manages student records with persistent file storage, providing reliable and easy-to-use operations for adding, updating, deleting, searching, and displaying student information.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by Thiranex India , confirming successful completion of the 4-weeks Thiranex Virtual Internship Program in C++ Programming, are attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

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